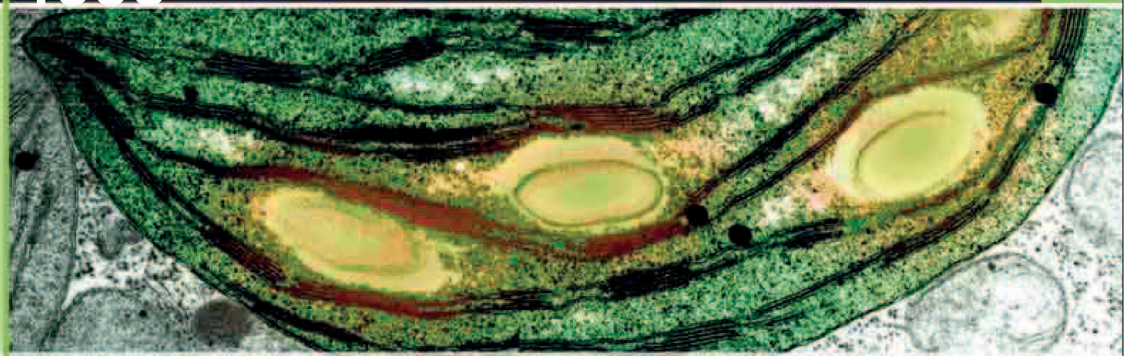
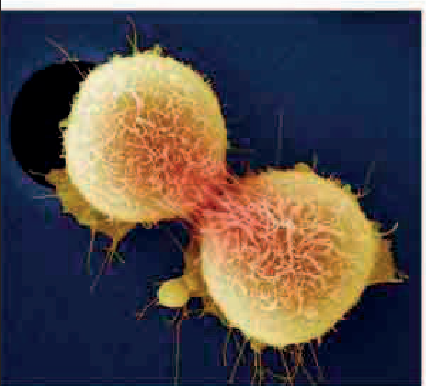
> **“** THE **SPACESUIT** STARTED **BEHAVING** ABSOLUTELY **DIFFERENT** FROM WHAT IT DID **ON THE GROUND. ”**
>
> **Alexey Leonov, Soviet cosmonaut,** 1965



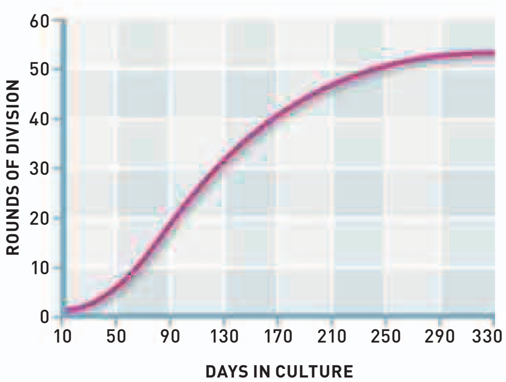
The endosymbiosis theory proposes that cellular organelles—such as the nucleus and chloroplasts shown here—once lived as independent organisms. Evidence for this comes in the form of their own functioning DNA.



Dividing cancer cells
Cancer cells have a genetic abnormality that makes them divide uncontrollably, deviating from the normal cellular aging process.

until the cells were no longer viable. Hayflick’s discovery had important implications in the **biology of cancer**. Cancer cells are abnormal in that their cell division is unchecked—so continuous division produces a tumor. A modern strategy in cancer treatment involves exposing the affected cells to drugs that encourage the natural corrosion of chromosomes.

In the 1960s, scientists **cracked the genetic code**. By 1961 it was known that DNA is a two-chain double helix that carries information for building proteins. The sequence of DNA chain building-blocks, called bases, determines the sequence of protein chain units, called amino acids. Between 1961 and 1966, cell biologists worked out the triplet combinations of bases that are encoded for all 20 different kinds of amino acids. The final link came in 1965, when American biochemist **Robert Holley** (1922–93) unraveled the **structure of tRNA** (transfer RNA)—the molecule that provided a physical link between DNA base code and assembling protein.

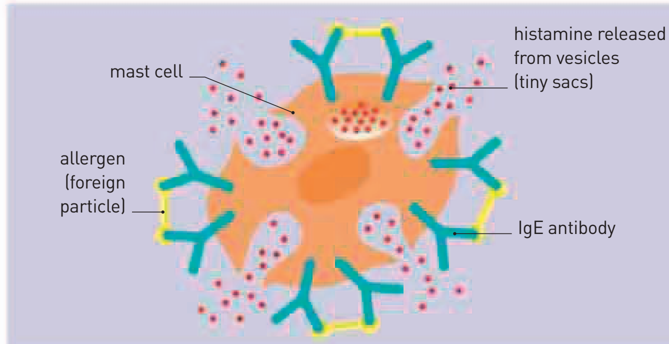


Hayflick’s graph of cell proliferation
Normal cells nurtured in the laboratory soon divide to produce a living culture—but after about 50 rounds of division, reproduction stops.

IN JUNE, after being rejected by more than a dozen scientific journals, a young scientist finally managed to publish a theory that would eventually revolutionize our understanding of the early history of life on Earth. American biologist **Lynn Margulis** (then married to Carl Sagan, see 1970) proposed that components of cells—such as the nucleus and chloroplasts—originally had independent lives. She suggested that millions of years ago bacteria-like life forms engulfed one another to form **the first complex cells, called eukaryotes**. Today, all animals, plants, and many microbes are made up of eukaryotic cells. Her **endosymbiotic theory** was initially resisted by most other scientists.

In July, Japanese husband and wife biologists **Kimishige** (b.1925) **and Teruko Ishizaka** (b.1926), working in the field of immunology, reported that they had discovered a **new class of antibodies**. These substances, called **immunoglobulin E (IgE)**, play a central role in making people oversensitive to certain triggers called allergens. Although they help in defending the body against certain kinds of parasites, IgE antibodies can also make it overproduce chemicals such as **histamine**, which triggers the massive inflammation associated with allergic reactions.

Meanwhile, in a Mt. Hotham Resort ski hut in Victoria, Australia, the discovery of a mouse-sized animal was causing a sensation among zoologists. This **mountain pygmy possum**—the only marsupial adapted to the snow-capped habitats of the Australian Alps—was previously known only from fossils and thought to be extinct for more than half a century. In 1966, the first living specimen was found.



ALLERGIC RESPONSE

IgE antibodies are the basis for allergic responses. When first exposed to an allergen (allergy-causing particle, such as pollen), white blood cells release IgE antibodies, which then bind to mast cells. When these IgE molecules bind to a second exposure of the same allergens, it makes the mast cell release histamine. This triggers the body’s allergic response symptoms.

Mountain pygmy possum
The mouse-sized alpine possum was discovered as a fossil in 1896, but a living animal was found in Australia in 1966.

- **March 18** Alexey **Leonov** becomes the first person to walk in space
- **March** American biochemist Robert Holley sequences **tRNA**—an intermediary of the generic code
- **March** Concept of **Hayflick limit** established
- **May** American astronomer Allan Sandage discovers radio-quiet **quasars**
- **June 8** Endosymbiotic theory of **eukaryotic cell origins** established
- **July** Japanese biologists Teruka and Kimishige Ishizaka discover **antibody class IgE**
- First living example of a **mountain pygmy possum** discovered in ski resort on Mt. Hotham, Australia